

Integrations

Mathematical tool which is used to add infinite parts of any quantity

$$(1) \int dx = x$$

$$(2) \begin{array}{|c|} \hline dA \\ \hline \end{array} \rightarrow A = \int dA = A$$

$$(3) \rightarrow \int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$\text{Ex } \int x^2 dx = \frac{x^3}{3}$$

$$\int x^{-2} dx = \frac{x^{-2+1}}{-1} = \frac{x^{-1}}{-1}$$

$$\textcircled{4} \rightarrow \int \text{constant} \cdot x^n dx = \text{constant} \int x^n dx$$

$$\text{Ex:} \int 3x^2 dx = 3 \int x^2 dx$$

$$= 3 \times \frac{x^3}{3}$$

$$= x^3$$

$$\text{Ex} \int -2m^4 dm = -2 \int m^4 dm$$

$$= -2 \times \frac{m^5}{5}$$

$$\textcircled{5} \rightarrow \int \text{constant} dx = \text{constant} \int dx$$

$$\int dx = x$$

$$\text{Ex.} \int 3 dx = 3 \int dx$$

$$= 3x$$

$$\int (m \pm n) dx = \int m dx \pm \int n dx$$

$$\textcircled{6} \Rightarrow \int (m \pm n) dx$$

$$= \int m dx \pm \int n dx$$

$$\text{Ex: } \int (x^2 + 4x) dx$$

$$= \int x^2 dx + \int 4x dx$$

$$\Rightarrow \frac{x^3}{3} + \frac{4x^2}{2}$$

$$\text{Ex: } \int (m^3 + m^2 + 7) dm$$

$$= \int m^3 dx + \int m^2 dx + \int 7 dx$$

$$\Rightarrow \frac{m^4}{4} + \frac{m^3}{3} + 7x$$

7:- With variables:-

$$i) \int \sin x \, dx = -\cos x$$

$$(ii) \int \cos x \, dx = \sin x$$

$$Ex: \int (\sin x + x^3) \, dx$$

$$= \int \sin x \, dx + \int x^3 \, dx$$

$$= \int -\cos x + \frac{x^4}{4}$$

8. Lower limit upper limit

$$\int_{\substack{\text{lower limit} \\ \text{final}}}^{\substack{\text{upper limit} \\ \text{3}}} x^2 \, dx = \left[\frac{x^3}{3} \right]_{\substack{\text{initial} \\ 0}}^{\substack{\text{final} \\ 3}}$$

$$Ex = \int_0^3 x^2 dx = \left[\frac{x^3}{3} \right]_0^3$$

$$= \left[\frac{x^3}{3} \right]_0^3$$

$$= \frac{(3)^3}{3} - \left(\frac{0^3}{3} \right) \rightarrow \text{lower}^{\text{final } x} - \text{initial } x$$

$$= \frac{27}{3} = 0 = 9$$

$$= \text{Ans} = 9.$$

$$Ex: \int_{-1}^{+1} 4x^3 dx$$

$$\Rightarrow 4 \int x^3 dx$$

$$= 4x \left(\frac{x^4}{4} \right)$$

$$= x^4$$

$$4x \left[\frac{x^4}{4} \right]_{-1}^{+1}$$

$$4x \left[\frac{(1)^4}{4} - \frac{(-1)^4}{4} \right]$$

$$= 4 \times 0 = 0$$

$$\therefore \text{Ans.} = 0$$

$$\text{Ex: } \int_0^{30^\circ} \sin x \, dx$$

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